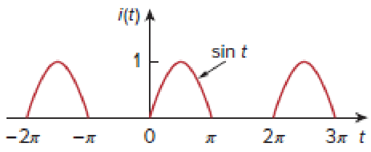
Chapter 17, Problem 55P

Problem

Obtain the exponential Fourier series expansion of the half-wave rectified sinusoidal current of Fig. 17.82.

Figure 17.82:



Step-by-step solution

Step 1 of 5

Refer to Figure 17.82 in the textbook.

The half wave rectified sinusoidal current is defined as,

$$i(t)=\begin{cases}\sin t & 0<t<\pi\\ 0 & \pi<t<2\pi\end{cases}$$

Consider the time period of $i(t)$ is $T=2\pi$.

Calculate the angular frequency, ω.

$$\begin{aligned}\omega&=\frac{2\pi}{T}\\&=\frac{2\pi}{2\pi}\\&=1\text{ rad/s}\end{aligned}$$

Step 2 of 5

Consider the general equation for exponential Fourier series.

$$i(t)=\sum_{n=-\infty}^{\infty}c_n e^{jn\omega t} \quad \ldots\ldots (1)$$

$$\begin{aligned}c_n&=\frac{1}{T}\int_0^T i(t)e^{-jn\omega t}dt\\&=\frac{1}{2\pi}\int_0^{2\pi} i(t)e^{-jnt}dt\\&=\frac{1}{2\pi}\left[\int_0^{\pi} i(t)e^{-jnt}dt+\int_{\pi}^{2\pi} i(t)e^{-jnt}dt\right]\\&=\frac{1}{2\pi}\left[\int_0^{\pi} \sin(t)e^{-jnt}dt+\int_{\pi}^{2\pi} (0)e^{-jnt}dt\right]\end{aligned}$$

Step 3 of 5

Consider,

$$c_n = \frac{1}{2\pi}\left[\int_0^{\pi} \sin(t) e^{-jnt} dt + \int_{\pi}^{2\pi} (0) e^{-jnt} dt\right]$$

$$c_n = \frac{1}{2\pi}\int_0^{\pi} \sin(t) e^{-jnt} dt \text{ ...... (2)}$$

Consider $I = \int \sin(t) e^{at} dt$.

Determine the value of I.

$$I = \int \sin(t) e^{at} dt$$

$$I = \frac{\sin t e^{at}}{a} - \int \frac{\cos t e^{at} dt}{a}$$

$$I = \frac{\sin t e^{at}}{a} - \left(\frac{\cos t e^{at}}{a^2} - \int \frac{\sin t\, e^{at}}{a^2} dt\right)$$

$$I + \frac{I}{a^2} = \frac{\sin t e^{at}}{a} - \frac{\cos t e^{at}}{a^2}$$

Thus,

$$I\frac{\left[1+a^2\right]}{a^2} = \frac{\sin t e^{at}}{a} - \frac{\cos t e^{at}}{a^2}$$

Step 4 of 5

Consider,

$$I\frac{\left[1+a^2\right]}{a^2} = \frac{\sin t e^{at}}{a} - \frac{\cos t e^{at}}{a^2}$$

$$I = \frac{e^{at}}{1+a^2}\left[a \sin t - \cos t\right]$$

Replace $a = -jn$ in equation.

$$I = \frac{e^{-jnt}}{1+(-jn)^2}\left[(-jn)\sin t - \cos t\right]$$

Replace I value in equation 2.

$$c_n = \frac{1}{2\pi}\left[\frac{e^{-jnt}}{1+(-jn)^2}\left[(-jn)\sin t - \cos t\right]\right]_0^{\pi}$$

$$= \frac{1}{2\pi}\left[\frac{e^{-jnt}}{1-n^2}\left[-\cos t - jn \sin t\right]\right]_0^{\pi}$$

$$= \frac{1}{2\pi}\left[\left[\frac{e^{-j\pi t}}{1-n^2}\left[-\cos \pi - jn \sin \pi\right]\right] - \frac{e^{-0}}{1-n^2}\left[-\cos 0 - jn \sin 0\right]\right]$$

$$= \frac{1}{2\pi\left(1-n^2\right)}\left[e^{-j\pi}\left(-(-1)-0\right) - 1\left(-1-0\right)\right]$$

Thus,

$$c_n = \frac{1}{2\pi\left(1-n^2\right)}\left[e^{-j\pi} + 1\right]$$

Step 5 of 5

Replace c_n in equation 1.

$$i(t)=\sum_{n=-\infty}^{\infty} c_n e^{jn\omega_o t}$$

$$i(t)=\sum_{n=-\infty}^{\infty} \frac{e^{-jn\pi}+1}{2\pi\left(1-n^2\right)} e^{jnt}$$

Therefore, the exponential Fourier series is $\boxed{\sum_{n=-\infty}^{\infty} \frac{e^{-jn\pi}+1}{2\pi\left(1-n^2\right)} e^{jnt}}$.